

Functional Lineale-valued Sets*

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Abstract

Schalk and de Paiva’s $FP\text{-Set}$ construction provides a small and uniform machine for producing models of linear logic, encompassing phase spaces, coherence spaces, and hypercoherences by equipping sets with values in an algebra of truth-values (a lineale) and using relations as morphisms. This note asks what happens when relations are replaced by structure-preserving *functions*. The answer is the category **LinSet**, which retains the linear structure supplied by the lineale—negation and modalities from a bare lineale, and symmetric monoidal closure once L has the meets needed to discharge a quantifier—while acquiring the limits and colimits absent from its relational counterpart. Indeed, whenever the lineale is complete, **LinSet** is complete and cocomplete, with its ordinary categorical structure computed on underlying sets, making the construction substantially richer categorically than the usual models of intuitionistic linear logic. The closing sections impose the H -set-like conditions on these objects, ask which of them are linear, and then separate what the construction owes to its base category from what it owes to the lineale.

1 Why functions

This paper modifies a well-known family of constructions in which a set is equipped with a map into truth-values. One starts with a set A and a map

$$\alpha: A \times A \longrightarrow \Omega,$$

and lets the algebraic structure of Ω determine what properties α may have. Boolean-valued sets provide one of the earliest examples: when Ω is a complete Boolean algebra, α may be regarded as a Boolean-valued equality predicate. Replacing the Boolean algebra by a complete Heyting algebra gives the H -sets of Fourman–Scott [15] and Higgs [17], while the corresponding locale-theoretic picture leads to sheaves [20]. More generally, replacing Ω by a quantale gives Q -valued sets, or Q -sets e.g. [6]. These constructions differ in the algebra carried by the truth values, but share the same basic idea: a set is equipped with a generalized equality or relation whose values lie in an ordered algebra.

The Schalk–de Paiva construction $FP\text{-Set}$ [8, 23] takes this basic idea into linear logic. It replaces the cartesian algebra of ordinary truth values by a lineale [18], namely a partially ordered monoid equipped with a residuated implication. The monoidal multiplication is not required to be idempotent, and this makes it possible for the truth-value object itself to carry genuinely linear structure. In the $FP\text{-Set}$ construction, the objects are again sets equipped with

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lineale-valued maps, but the morphisms are relations rather than functions. Thus $F_P\text{-}\mathbf{Set}$ can be regarded as another member of this broad family of truth-value-valued set constructions, adapted to produce models of linear logic. In $F_P\text{-}\mathbf{Set}$ a morphism $(A, \alpha) \rightarrow (B, \beta)$ is a relation $R \subseteq A \times B$ satisfying

$$aRb, a'Rb' \implies \alpha(a, a') \leq \beta(b, b').$$

Here we make just one change: we replace relations by structure-preserving *functions*. Thus a morphism in $\mathbf{LinSet}$ is a function $f: A \rightarrow B$ such that

$$\alpha(a, a') \leq \beta(fa, fa').$$

The objects, the lineale, and the pointwise definitions are otherwise unchanged. The question of this paper is whether this apparently small change produces a useful category, and what becomes of the linear structure when $\mathbf{Rel}$ is replaced by $\mathbf{Sets}$.

The answer is that the change makes the source of the linear structure explicit, and that what makes it explicit also buys structure the relational construction cannot have. $\mathbf{Sets}$ contributes no linear structure of its own, so every linear feature of $\mathbf{LinSet}$ must come from the lineale: monoidal closure from residuation, negation from a chosen d , modality from a storage operator, duplicable objects those with idempotent values below the unit. And because $\mathbf{Sets}$ is not self-dual, limits and colimits are no longer forced to coincide: the terminal object is genuinely terminal, equalizers and coequalizers exist, the forgetful functor is topological, and $\mathbf{LinSet}$ is complete and cocomplete with limits and colimits computed on carriers. Models of intuitionistic linear logic do not usually have this combination. These are not all bought at the same price: what each of them asks of L — a bare lineale, a lattice with top and bottom, or completeness — is set out in Section 2.2.

The comparison with Boolean-valued sets, H -sets, sheaves, and Q -sets also suggests a use for this additional structure. In those constructions, conditions imposed on the truth-value-valued relation—such as symmetry, transitivity, boundedness, and local identity—turn a general relation into an analogue of equality and allow one to construct richer categorical structures, including sheaf, tripos, and topos-like constructions. The H -set and Q -set literature therefore provides a natural question for the linear setting. Starting with a lineale-valued relation in $\mathbf{LinSet}$, which of these equality-like conditions remain compatible with genuinely linear structure? What are the appropriate linear analogues of H -sets or Q -sets? And can such objects support a construction playing the role that the tripos-to-topos construction plays in the intuitionistic setting?

Section 5 answers the first two of these questions and closes off one obvious route to the third. The conditions ask that similarity be symmetric and transitive, that it be bounded by existence, and that existence act as an identity for it. The first three survive the passage to a lineale; the fourth does not, and it is where all the difficulty is. It requires that the extent of a point be idempotent, so that however partial a point's existence may be, that existence behaves classically. The second question is answered along the way: the analogue does exist, its objects being those whose existence takes values in the idempotent part of L while their similarities stay unconstrained, and Example 5.3 shows the class is not degenerate. The third it cannot support. Our reading is that the difficulty lies in the extent itself, in taking self-similarity as the measure of how much a point exists, and that a linear analogue would need some notion of linear extent, which we do not have. What we settle instead is the question the paper opened with: with functions in place of relations, the contribution of the lineale can be told apart from what the relational construction inherited from $\mathbf{Rel}$.

2 The construction **LinSet**

The Schalk–de Paiva F_P -**Set** construction gives not a single model of linear logic but a family of them: symmetric monoidal closed with a linear exponential comonad, hence a model of intuitionistic linear logic, and $*$ -autonomous, hence of the classical one, whenever the poset supplies a dualizing element. Several of the standard models of linear logic are instances of it: phase spaces, coherence spaces, hypercoherences. In that construction, an object is a set A with a map $\alpha: F(A) \rightarrow P$, for F an endofunctor on **Sets** and P a poset. Throughout this paper F is the diagonal¹ $\Delta: A \mapsto A \times A$ and P is a lineale L , so that an object is a set A together with a map $\alpha: A \times A \rightarrow L$ and nothing else. We start by recalling what a *lineale* is.

Definition 2.1 (lineale). A *lineale* is a partially ordered commutative monoid $(L, \leq, \cdot, 1)$ in which multiplication is monotone and has a right adjoint: for each b there is $b \multimap (-)$ with $a \cdot b \leq c$ if and only if $a \leq b \multimap c$.

Here are three simple examples of lineales to keep in mind. A *locale*, where $\cdot$ is $\wedge$, and everything is idempotent; the *Lawvere quantale* $([0, \infty], \geq, +, 0)$, where the order is reversed, and nothing but 0 is idempotent; and the three-element *Kleene chain*, which is a Heyting algebra and not a Boolean one. The second is where the construction is most linear, and the third is small enough to compute.

Then the main definition of this note:

Definition 2.2 (lineale-valued set). Let L be a lineale. A *lineale-valued set* is a pair (A, α) with A a set and $\alpha: A \times A \rightarrow L$ a function. No conditions are imposed on the function α .

The intended reading of the function $\alpha(a, a')$ is the extent to which a and a' are identified, so that a morphism is a map that does not separate what was identified. Two things follow from the absence of axioms on α . First, since α need not be symmetric or transitive, an object is not an L -valued equivalence; some conditions or axioms are added in Section 5, and it matters which results are still true without them. Second, on the diagonal, $\alpha(a, a)$ is unconstrained but carries information: it is the *extent* of a , the degree to which a exists, and it is what the boundedness and local identity axioms of the Q -set literature are about.

Definition 2.3. A morphism $(A, \alpha) \rightarrow (B, \beta)$ of **LinSet** is a function $f: A \rightarrow B$ such that

$$\alpha(a, a') \leq \beta(f(a), f(a')) \quad \text{for all } a, a' \in A.$$

Composition and identities are those of **Sets**.

That **LinSet** is a category needs no argument: the condition composes by transitivity of $\leq$ and holds for the identity by reflexivity. Nothing beyond the partial order on L is used, and in particular neither $\cdot$ nor $\multimap$ of the lineale appears so far.

Remark 2.4 (The forgetful functor). Forgetting α gives a functor $U: \mathbf{LinSet} \rightarrow \mathbf{Sets}$, and two things about it are immediate. It is faithful, a morphism being a function with a property. And if L has a bottom and a top it has both a left and a right adjoint, $A \mapsto (A, \perp)$ and $A \mapsto (A, \top)$: a function $A \rightarrow U(B, \beta)$ is a morphism $(A, \perp) \rightarrow (B, \beta)$ since $\perp \leq \beta$ always, and a function $U(B, \beta) \rightarrow A$ is a morphism $(B, \beta) \rightarrow (A, \top)$ since $\beta \leq \top$ always, both bijections being the identity on underlying functions. So every set carries a free and a cofree lineale-valued structure.

¹A generalisation in the F direction might be available but is not pursued.

2.1 Examples of lineale-valued sets

We now give some examples of categories of lineale-valued sets. First, the historical precedent:

Example 2.5 (Locale-valued sets). Take L a locale. An object (A, α) is a set A with an L -valued similarity $\alpha : A \times A \rightarrow L$, and a morphism is a function that does not decrease it. Imposing symmetry and transitivity gives the Ω -sets of the sheaf-theoretic tradition [15], minus their completeness requirements; the locale case returns in Section 5 as the test any completion would have to pass. This is also the older sense of “ L -set”, which is locale-valued; ours are lineale-valued, the value object need not be idempotent, and dropping idempotence is precisely what makes the resulting logic linear. We say “lineale-valued set” throughout.

Example 2.6 (Generalised metric spaces). Take L the Lawvere quantale $([0, \infty], \geq, +, 0)$, whose order is that of the reals reversed. Writing $d = \alpha$, the condition of Definition 2.3 reads $d(fa, fa') \leq d(a, a')$. So an object is a set with a distance function subject to no axioms, and a morphism is a non-expansive map. In the vocabulary of Section 5, transitivity (T) is the triangle inequality and $E_x = 1$ is $d(x, x) = 0$; imposing those two gives exactly the generalised metric spaces of Lawvere [21] and their short maps. Note that symmetry (S) is not imposed there — it is the axiom Lawvere drops — so his spaces are the (T)-and-reflexive objects and not the H -set-like ones of Definition 5.1.

Finally, some important but degenerate cases:

Example 2.7 (Degenerate cases). If L is the one-element lineale $\mathbf{1}$ then every function is a morphism and $\mathbf{LinSet}$ is $\mathbf{Sets}$. If $L = \mathbf{2}$ then an object is a set with a binary relation and a morphism a relation-preserving function, so $\mathbf{LinSet}$ is the category of directed graphs; the linear structure of Section 3 is then the familiar one on graphs.

The first of these examples does make $\mathbf{LinSet}$ a degenerate model of linear logic, a cartesian closed category, instead of a monoidal closed one. The corresponding statement for the Dialectica construction [7], one of the intuitions for the Schalk–de Paiva construction, is false. An object of $\mathbf{LinSet}$ has one carrier, and both its tensor and its product are carried by $A \times B$, differing only in whether the values are combined by the tensor $\cdot$ or by $\wedge$ (Sections 4 and 3). So when L is a locale, the two coincide, the internal hom becomes the function space, and $\mathbf{LinSet}$ is cartesian closed — Proposition 4.4 states this precisely, once both structures have been defined. Nothing of the kind happens for the Dialectica construction. Over $\Omega = 2$, as classical a value object as there is, the construction [7] still yields a symmetric monoidal closed category and not a cartesian closed one. The linearity there is manufactured by the shape of the objects and morphisms, not inherited from the value object directly, which is exactly why the construction is interesting over $\mathbf{Sets}$ at all.

A non-example The Q -sets of the Alvim–Mendes–Mariano tradition [2, 3] impose four axioms on α — symmetry, transitivity, boundedness and local identity — and come in two versions, one with functional morphisms and one with Q -valued relational ones. The axioms are not theirs alone: they descend from the Ω -set axioms of [15], and the quantale-valued theory they belong to goes back to Höhle [16]. A quantale is a lineale, so one might expect $Q\text{-Set}$ to be $\mathbf{LinSet}$ over Q . It is not. $\mathbf{LinSet}$ takes the axiom-free objects with the functional morphisms, so the Q -sets are cut out of it by the four axioms rather than produced by it, and agrees with the Schalk–de Paiva relational lineale-valued category on objects but not on morphisms.

2.2 Why not a quantale?

Completeness of the lineale L is not generally assumed here. The literature usually asks for a quantale [22] — a complete lattice whose monoid preserves joins in each argument — and most of what follows does not need one. For a lineale the two halves of that definition are one condition rather than two: residuation makes $\cdot$ a left adjoint in each argument, so it preserves every join that exists, and a poset with all joins has all meets. A complete lineale is therefore already a commutative unital quantale, and conversely such a quantale is a lineale, its residuation given by the adjoint functor theorem for posets. So the only hypothesis at issue below is completeness, and the question is which results need it.

- A bare lineale suffices for the category itself, for the tensor, for the negation, for the modality and for the comonoid condition (Proposition 3.12). Composition and the coherence condition involve no joins or meets at all; the tensor uses the multiplication of the lineale; the negation is residuation against a chosen element of L ; the modality needs a storage operator on L . The equalizer and the initial object cost nothing, being the constructions of **Sets** with α restricted or empty.
- Lattice conditions, short of completeness, buy the rest of the additives and the exponential isomorphisms. Binary meets give binary products; a top gives the terminal object, which is the additive truth; the two together give Girard’s isomorphisms (Proposition 3.13), whose second equation is about $\top$; a bottom gives binary coproducts. So a lattice with top and bottom yields all finite limits, and all finite colimits except coequalizers.
- Three constructions need completeness, and for the same reason: each formula is indexed by something arbitrary. The internal-hom takes a meet over $A \times A$, $\bigwedge_{a,a'} [\alpha(a,a') \multimap \beta(ua,va')]$; the coequalizer of f,g takes a join over the members of a pair of classes, $\bigvee \beta(c,c')$; and the initial lift of a family f_i takes a meet over that family, $\bigwedge_i \beta_i(f_i a, f_i a')$. Carriers, classes and families being arbitrary, these want arbitrary meets and joins.

So the hypotheses are a lineale throughout; a lattice with top and bottom for the additives and the finite limits; and completeness for the internal hom, for coequalizers, and for the topological property that yields all small limits and colimits. Restricting to finite carriers brings the first two of those three down to finite meets and joins, with a top and a bottom for the empty ones; the third is indexed by a family rather than by a carrier, so finiteness of the carriers does nothing for it. We say “quantale” when the standing hypothesis is completeness and “lineale” otherwise, instead of silently assuming the stronger hypothesis.

3 Monoidal closure, negation, modality

The multiplicative exponential structure of **LinSet** is the main reason for writing this note. Everything in it is a statement about L , transported pointwise, and **Sets** contributes only the carrier sets: the tensor is carried by the cartesian product of carriers, the internal hom by the set of all functions, and what makes them linear is the multiplication and the residuation of the lineale; only the internal hom asks for anything further, namely the meets of Section 2.2. One observation deals with coherence before we start.

Remark 3.1 (Faithfulness and coherence). The forgetful functor of Remark 2.4 is faithful, and that deals with coherence throughout this paper. Every structural morphism here — associators,

unitors, symmetries, the counit and comultiplication of the modality, the canonical map into the double negation — is carried by a function of **Sets**, usually an identity function. Any coherence diagram between such morphisms therefore commutes in **LinSet** if and only if it commutes in **Sets**, where it does. So no coherence condition below requires checking: the content of each proposition is entirely that the functions concerned are morphisms, that is, that the corresponding inequality holds in L .

3.1 The monoidal structure

We first define the tensor product inherited from the lineale monoidal structure.

Definition 3.2 (tensor). For lineale-valued sets (A, α) and (B, β) over L put

$$(A, \alpha) \otimes (B, \beta) := (A \times B, \alpha \otimes \beta), \quad (\alpha \otimes \beta)((a, b), (a', b')) := \alpha(a, a') \cdot \beta(b, b'),$$

with unit $I := (\{*\}, \tau)$, $\tau(*, *) := 1$, where $\cdot$ is the monoidal structure of L .

Proposition 3.3. (**LinSet**, $\otimes, I$) is a symmetric monoidal category, using only the monoid structure of L and its monotonicity.

Proof. On morphisms, $f \otimes g$ is $f \times g$, which is a morphism because $\alpha(a, a') \leq \alpha'(fa, fa')$ and $\beta(b, b') \leq \beta'(gb, gb')$ give $\alpha(a, a') \cdot \beta(b, b') \leq \alpha'(fa, fa') \cdot \beta'(gb, gb')$ by monotonicity of $\cdot$ in each argument. The associator, unitors and symmetry are those of **Sets**, and each is a morphism in both directions because the corresponding equation holds in the monoid — associativity, unitality with 1, commutativity — and their coherence is inherited from **Sets** by Remark 3.1. $\square$

Definition 3.4 (internal hom). Suppose L has all meets; the ones used are indexed by $A \times A$. Put

$$[(A, \alpha), (B, \beta)]_{\mathbf{LinSet}} := (B^A, [\alpha, \beta]), \quad [\alpha, \beta](u, v) := \bigwedge_{a, a' \in A} [\alpha(a, a') \multimap \beta(u(a), v(a'))],$$

the carrier being all functions $A \rightarrow B$, not only the morphisms of **LinSet**, and u, v ranging over them.

The morphism condition of Definition 2.3 is a statement about all pairs (a, a') , and in Definition 3.4 that quantifier has to be discharged inside L , since there is nowhere else to put it: the internal hom must be a single object with a single structure map. This is where completeness first appears, and it is the domain that fixes what is needed. The elements of B supply values but not indices, so a finite A asks only for finite meets however large B is. This immediately gives the monoidal *closed* structure.

Proposition 3.5 (**LinSet** is monoidal closed). Suppose L has the meets of Definition 3.4. Then **LinSet** is symmetric monoidal closed: there is a bijection $\text{Hom}(C \otimes A, B) \cong \text{Hom}(C, [A, B])$, natural in each variable.

Proof. A morphism $(C, \gamma) \otimes (A, \alpha) \rightarrow (B, \beta)$ is a function $h: C \times A \rightarrow B$ with

$$\gamma(c, c') \cdot \alpha(a, a') \leq \beta(h(c, a), h(c', a')) \quad \text{for all } c, c', a, a'.$$

By residuation this is $\gamma(c, c') \leq \alpha(a, a') \multimap \beta(h(c, a), h(c', a'))$ for all a, a' , and since the left side does not depend on a, a' it is equivalent to $\gamma(c, c') \leq \bigwedge_{a, a'} [\alpha(a, a') \multimap \beta(\hat{h}(c)(a), \hat{h}(c')(a'))]$, which

is the condition on the curried $\hat{h}: C \rightarrow B^A$. Currying is a bijection on the underlying functions and the two conditions correspond, so it restricts to a bijection on morphisms; naturality is naturality of currying in **Sets**. $\square$

The carrier of $[A, B]_{\mathbf{LinSet}}$ is all of B^A , not only the morphisms, and its structure map is binary like every α : it compares two functions, one in each argument. Taking both to be the same f gives the value at (f, f) , and f is a morphism exactly when that value is above 1, since $1 \leq \alpha(a, a') \multimap \beta(fa, fa')$ says $\alpha(a, a') \leq \beta(fa, fa')$. So the morphisms $A \rightarrow B$ are the points $I \rightarrow [A, B]$, which is Proposition 3.5 at $C = I$ and the usual reading of an internal hom in an enriched setting. As in any monoidal closed category, $[-, -]$ is a functor $\mathbf{LinSet}^{\text{op}} \times \mathbf{LinSet} \rightarrow \mathbf{LinSet}$; concretely, for $f: A \rightarrow A'$ the map $[f, B]: [A', B] \rightarrow [A, B]$ is precomposition with f , and it is a morphism because $\alpha(a, a') \leq \alpha'(fa, fa')$ makes each meetand for A larger than the corresponding one for A' .

3.2 Linear Negation

The tensor was an operation of L applied pointwise, with coherence coming for free; the internal hom needed more, both a meet to discharge the quantifier and a proof that the result is right adjoint to the tensor. Negation is pointwise again, and in that respect is like the tensor. What is new is that it does not stay inside the category.

Linear negation is not primitive: in a $*$ -autonomous category it is $A \multimap D$ for a chosen dualizing D , and the corresponding operation on values is $q \mapsto q \multimap d$ for a chosen d in L . We keep the carrier, so that η_A below is the identity function and involutivity becomes an equation in L rather than a bijection of sets; taking $A^\perp := [A, D]$ for a fixed object D instead would replace the carrier A by D^A , so that $A^{\perp\perp}$ has carrier $D^{(D^A)}$: the canonical map would be a function $A \rightarrow D^{(D^A)}$ rather than an identity, and for $|D| \geq 2$ no such map is a bijection. Involutivity would be out of reach for reasons having nothing to do with L . What breaks is the direction. Relations can be reversed and functions cannot, so where the relational construction of [23] obtains a contravariant endofunctor for free, here the negation reverses the order on the values and lands in **LinSet** over L^{op} .

Definition 3.6 (negation). Let $d \in L$. Put $(A, \alpha)^\perp := (A, \alpha^\perp)$ with $\alpha^\perp(a, a') := \alpha(a, a') \multimap d$.

With the definition in hand, one question is whether anything of the classical negation survives. One half does, for free: residuation gives $q \leq (q \multimap d) \multimap d$ in any lineale, so $\alpha \leq \alpha^{\perp\perp}$ pointwise and the identity function is always a morphism into the double negation. The other half is a condition on L and d , not a theorem. So the negation is semi-involutive, as in FILL [13], where the canonical map exists and the reverse inequality is exactly what fails.

Proposition 3.7 (duality). *The identity function is a morphism $\eta_A: (A, \alpha) \rightarrow (A, \alpha)^{\perp\perp}$, and it is an isomorphism if and only if $(q \multimap d) \multimap d = q$ for every q in the image of α .*

Proof. $q \leq (q \multimap d) \multimap d$ is the unit of the residuation adjunction, so $\alpha \leq \alpha^{\perp\perp}$ pointwise, which is the condition on the identity function. It is an isomorphism exactly when the inequality is an equality: the identity on (A, α) is the identity function, so an isomorphism in **LinSet** is a bijection with α preserved in both directions, and there is no coarser notion available to absorb the gap. $\square$

The definition so far acts only on objects, and a connective has to act on morphisms too. Here the pointwise formula settles it, but not in the shape one expects: residuation is antitone in its first argument, so a morphism f with $\alpha \leq \beta \circ (f \times f)$ becomes a function satisfying $\alpha^\perp \geq \beta^\perp \circ (f \times f)$ — the same function, the inequality reversed. That is a morphism of the category built over L^{op} , not of **LinSet** itself. We write **LinSet** $_L$ for **LinSet** when the base needs naming.

Proposition 3.8. $(-)^{\perp}$ is a covariant functor **LinSet** $_L \rightarrow \mathbf{LinSet}_{L^{\text{op}}}$, carried by the identity on underlying functions. Consequently $(-)^{\perp\perp}$ is a covariant endofunctor of **LinSet** $_L$ and η is a natural transformation $\text{id} \Rightarrow (-)^{\perp\perp}$.

Proof. Let $f: (A, \alpha) \rightarrow (B, \beta)$ in **LinSet** $_L$, so $\alpha \leq \beta \circ (f \times f)$ pointwise in L . Residuation is antitone in its first argument, so $\alpha^\perp \geq \beta^\perp \circ (f \times f)$, which is exactly the condition for f to be a morphism $(A, \alpha)^\perp \rightarrow (B, \beta)^\perp$ of **LinSet** $_{L^{\text{op}}}$. The same computation run in L^{op} takes a morphism there to one of **LinSet** $_L$, so the composite is an endofunctor of **LinSet** $_L$. Functoriality and the naturality of η are then statements about identity functions in **Sets**, by Remark 3.1. $\square$

The negation is functorial; it is not an operation within a single category. It takes **LinSet** $_L$ to **LinSet** $_{L^{\text{op}}}$, which has the same objects and the same underlying functions but the reversed inequality as its coherence condition. That is the description of a same-carrier negation over a value object whose morphisms cannot be reversed, and it is why $(-)^{\perp\perp}$ is the operation that acts on **LinSet** $_L$ itself. Note that L^{op} need not be a lineale — residuation may fail — but **LinSet** $_{L^{\text{op}}}$ is defined regardless, the category asking only for the order.

If d is dualizing, so that $(q \multimap d) \multimap d = q$ throughout, then $q \mapsto q \multimap d$ is an order-isomorphism $L \cong L^{\text{op}}$; it therefore induces an isomorphism **LinSet** $_{L^{\text{op}}} \cong \mathbf{LinSet}_L$, and under that identification $(-)^{\perp}$ becomes an involutive endofunctor of a single category. So the two conditions that one would want of a negation — being an endofunctor and being involutive — arrive together and for the same reason, and Proposition 3.7 is the statement that neither is automatic.

What does depend on the negation is the comparison with $\ast$ -autonomy: $\ast$ -autonomy in this setting is exactly the statement that d is dualizing, a condition on L and d and not something the passage to functions forecloses. It does not follow that there is no $\wp$. We do not want one defined by de Morgan from $\otimes$ and $(-)^{\perp}$: in intuitionistic logic conjunction, disjunction and implication are independent connectives, and the linear analogue should be too, with $\wp$ given by its own monoidal structure and related to $\otimes$ by the two linear distributivities rather than by a negation. That is the shape of Full Intuitionistic Linear Logic [13], where both distributivities hold and the negation is not involutive, and it is the shape we expect here. The transport is immediate, everything in this section being pointwise.

Proposition 3.9 (linear distributivity). *Suppose L carries a second commutative monotone monoid $(\wp, \perp_{\wp})$ satisfying the two linear distributivities*

$$x \cdot (y \wp z) \leq (x \cdot y) \wp z, \quad (x \wp y) \cdot z \leq x \wp (y \cdot z).$$

Put $(\alpha \wp \beta)((a, b), (a', b')) := \alpha(a, a') \wp \beta(b, b')$ on the carrier $A \times B$, with unit $(\{\}, \perp_{\wp})$. Then $(\mathbf{LinSet}, \wp)$ is symmetric monoidal and the two linear distributivities hold in **LinSet**, so that $(\mathbf{LinSet}, \otimes, \wp)$ is a linearly distributive category in the sense of Cockett and Seely [9].*

Proof. That $\mathfrak{V}$ is a symmetric monoidal structure is the proof of Proposition 3.3 with $\cdot$ replaced by $\mathfrak{V}$, using only commutativity, associativity, unitality and monotonicity. For the distributivities, the map $A \otimes (B \mathfrak{V} C) \rightarrow (A \otimes B) \mathfrak{V} C$ is carried by the associator of **Sets**, and the condition on it at $((a, (b, c)), (a', (b', c')))$ is

$$\alpha(a, a') \cdot (\beta(b, b') \mathfrak{V} \gamma(c, c')) \leq (\alpha(a, a') \cdot \beta(b, b')) \mathfrak{V} \gamma(c, c'),$$

which is the first hypothesis; the second distributivity is symmetric. Their coherence is that of the associators in **Sets**, by Remark 3.1. $\square$

3.3 The modality

Everything so far belongs to the multiplicative fragment, and a symmetric monoidal closed category (even with a $\mathfrak{V}$ operator) models that fragment and no more. What makes linear logic a logic rather than a fragment of one is that the structural rules return in marked form: contraction and weakening are not abolished but licensed where a formula is duplicable and erasable, and it is the modality that does the licensing — which is also what lets intuitionistic implication be read as $!A \multimap B$. So the **LinSet** construction owes us a modality.

It comes under the same discipline as the rest. The tensor came from the multiplication of L , the internal hom from its residuation, the negation from residuation against a chosen element; the modality comes from an operator on L , applied pointwise. On a poset a comonad is an interior operator, monotone with $!q \leq q$ and $!q \leq !!q$, and that alone yields dereliction and promotion. The two further conditions below, $!q \leq 1$ and $!q \cdot !q = !q$, are the algebraic form of erasability and duplicability at a marked formula; an IL-algebra carrying such an operator is what Troelstra calls an algebra with storage [12]. So what we ask of L is a storage operator, and the modality on **LinSet** is that operator composed with α . The relational category inherits a modality $!$ from **Rel** whatever L does, so its modality reports nothing about the value object; here there is a modality exactly when L supplies one.

Definition 3.10 (modality $!$). Suppose L carries a storage operator $!$: a monotone map with $!q \leq q$, $!q \leq 1$, $!q \leq !!q$ and $!q \cdot !q = !q$. Put $!(A, \alpha) := (A, ! \circ \alpha)$.

Proposition 3.11. *The operator $!$ is a linear exponential comonad on **LinSet**, with counit and comultiplication carried by identity functions.*

Proof. Functoriality: $\alpha \leq \beta \circ (f \times f)$ gives $!\alpha \leq !\beta \circ (f \times f)$ by monotonicity of $!$. The counit $!A \rightarrow A$ is the identity function, whose condition is $!\alpha \leq \alpha$, that is dereliction; the comultiplication $!A \rightarrow !!A$ likewise, with condition $!\alpha \leq !!\alpha$. The comonad equations are equations between identity functions and hold by Remark 3.1. $\square$

Proposition 3.12 (comonoids). *The diagonal $\Delta: A \rightarrow A \otimes A$, $a \mapsto (a, a)$, is a morphism if and only if $\alpha(a, a') \leq \alpha(a, a') \cdot \alpha(a, a')$ for all a, a' ; and $A \rightarrow I$ is a morphism if and only if $\alpha(a, a') \leq 1$. Hence $!A$ is a comonoid for every A , and every object is one when L is a locale.*

Proof. The two conditions are Definition 2.3 applied to Δ and to the unique function $A \rightarrow \{*\}$, given Definition 3.2. For $!A$ they read $!\alpha \leq !\alpha \cdot !\alpha$ and $!\alpha \leq 1$, which are idempotence and weakening. For a locale, $\cdot$ is $\wedge$ and 1 is the top, so both hold for any α . $\square$

Two isomorphisms due to Girard relate the modality to the additives. They hold here exactly when the corresponding equations hold in L .

Proposition 3.13. *Suppose L has binary meets and satisfies $!(x \wedge y) = !x \cdot !y$ and $!\top = 1$. Then $!(A \times B) \cong !A \otimes !B$ and $!T \cong I$, both carried by identity functions, where $A \times B$ and T are the product and terminal object of Proposition 4.1.*

Proof. $!(A \times B)$ has carrier $A \times B$ and values $!(\alpha(a, a') \wedge \beta(b, b'))$, while $!A \otimes !B$ has the same carrier and values $!\alpha(a, a') \cdot !\beta(b, b')$. The identity function is a morphism in each direction precisely when those two agree, which is the first hypothesis applied pointwise. Similarly $!T$ and I both have carrier $\{*\}$, with values $!\top$ and 1 , and the second hypothesis makes the identity a morphism both ways. By Remark 3.1 nothing further is needed. $\square$

Two things are worth drawing out. First, the modality is not imported. The relational construction inherits its $!$ from **Rel** unconditionally, which is convenient and tells one nothing about L ; this one is L 's own storage operator applied pointwise, so it exists exactly when L says it does and reflects the lineale one is working over, Proposition 3.13 being the sharpest case: Girard's isomorphisms hold exactly when the corresponding equations hold in L .

Second, Proposition 3.12 says the duplicable/erasable objects are exactly those whose values are idempotent and below the unit and Proposition 3.11 says the modality produces them. So $!$ marks the duplicable/erasable objects rather than making everything duplicable/erasable, which is what one wants of it, and the degeneracy of the locale case (Proposition 4.4) is the case where the marking is vacuous because everything was duplicable/erasable already.

4 Limits and colimits

Most of what follows is about the logic. The product and the coproduct are the additive conjunction and disjunction, $\&$ and $\oplus$; the terminal object is the additive truth $\top$, the unit of $\&$; and the empty coproduct is $\mathbf{0}$, the unit of $\oplus$. Together with the multiplicatives of Section 3 and the modality, this interprets every² connective of intuitionistic linear logic.

The equalizer and the coequalizer are not logical connectives, and we discuss them because the category is the object of study in this section. They are also where the relational construction fails. Everything here is the construction from **Sets** with α carried along, and the content is that this carrying along works. We give the finite cases explicitly, since the formulas are what one calculates with, and note in each case which fragment of L is used.

Proposition 4.1 ((co)-products, (co)-equalizers). *Let (A, α) , (B, β) be lineale-valued sets and $f, g: (A, \alpha) \rightarrow (B, \beta)$ morphisms.*

1. *If L has binary meets, $(A \times B, \alpha \wedge \beta)$ is the product, where $(\alpha \wedge \beta)((a, b), (a', b')) = \alpha(a, a') \wedge \beta(b, b')$. If L has a top, $T = (\{*\}, \top)$ is the terminal object.*
2. *If L has a bottom, $(A \sqcup B, \alpha \sqcup \beta)$ is the coproduct, where $\alpha \sqcup \beta$ is α on A , β on B and $\perp$ across. The object $(\emptyset, \emptyset)$ is the initial object, using nothing of L .*
3. *$(\{a : f(a) = g(a)\}, \alpha)$ with the inclusion is the equalizer, using nothing of L .*
4. *If L has the relevant joins, $(B/\sim, \gamma)$ with the quotient map is the coequalizer, where $\sim$ is the equivalence relation generated by $f(a) \sim g(a)$ and $\gamma([b], [b']) = \bigvee_{c \in [b], c' \in [b']} \beta(c, c')$.*

²To deal with $?$ use the negation, as $(!(A^\perp))^\perp$, whenever d is dualizing.

Proof. (1) A meet is below each factor, so the projections are morphisms, and for $h = \langle h_1, h_2 \rangle$ from (C, γ) the condition $\gamma \leq (\alpha \wedge \beta) \circ (h \times h)$ holds if and only if it holds for each factor; the condition on a function into T is $\alpha(a, a') \leq \top$, which holds. (2) Dually, $\perp$ is below everything, so the injections are morphisms and the condition on a cotupling splits into the conditions on the two components, the cross terms being satisfied by $\perp$; the unique function out of $\emptyset$ is vacuously a morphism. (3) The inclusion preserves α on the nose, and a morphism h with $fh = gh$ factors through the subset as a function, the factorisation being a morphism because α there is the restriction. (4) The quotient map is a morphism because $\beta(b, b')$ is one of the terms of the join defining $\gamma([b], [b'])$; and given h with $hf = hg$, the induced k on classes satisfies $\beta(c, c') \leq \delta(hc, hc') = \delta(k[b], k[b'])$ for every term of that join, since h is constant on classes, so the join is below $\delta(k[b], k[b'])$. $\square$

Corollary 4.2. *If L is a lattice with top and bottom, **LinSet** has finite products, coproducts and equalizers, hence all finite limits.*

The constructions above have the same shape, and there is one reason for it. Take a family of functions $f_i: A \rightarrow B_i$ into the carriers of lineale-valued sets (B_i, β_i) . Each f_i is a morphism precisely when $\alpha \leq \beta_i \circ (f_i \times f_i)$, so the constraints are all upper bounds, and the largest α meeting them at once is the pointwise meet $\bigwedge_i \beta_i(f_i a, f_i a')$. That choice is *initial*: a function into (A, α) is a morphism exactly when each of its composites with the f_i is. A faithful functor for which every such family has a unique initial lift is called *topological* [1], and topological functors create limits and colimits, so all of them come at once, by taking them in **Sets** and lifting.

Proposition 4.3. *If L is complete then the forgetful functor $U: \mathbf{LinSet} \rightarrow \mathbf{Sets}$ is topological: every small family $f_i: A \rightarrow U(B_i, \beta_i)$ has a unique initial lift, namely $\alpha(a, a') = \bigwedge_i \beta_i(f_i a, f_i a')$, and dually.*

Proof. U is faithful by Remark 2.4. For the initial lift, α as displayed makes each f_i a morphism, and a function $g: (C, \gamma) \rightarrow (A, \alpha)$ is a morphism if and only if $\gamma \leq \beta_i \circ (f_i g \times f_i g)$ for every i , which is the defining property of the meet; uniqueness is uniqueness of the meet. $\square$

LinSet is complete and cocomplete whenever L is, with everything computed on carriers. The abstract reason and the concrete failures below agree: a topological functor needs initial lifts to be *unique*, and uniqueness is exactly what multi-valuedness of relations destroys.

Proposition 4.4. *If L is a locale then $\otimes$ is the categorical product, I is the terminal object, and **LinSet** is cartesian closed.*

Proof. In a locale $\cdot$ is $\wedge$ and 1 is $\top$. So the tensor of Definition 3.2 is carried by $A \times B$ with values $\alpha(a, a') \wedge \beta(b, b')$, which is the product of Proposition 4.1(1); and the unit $I = (\{*\}, 1) = (\{*\}, \top)$ is the terminal object of the same item. A locale has all meets, so the internal hom of Definition 3.4 exists, and Proposition 3.5 makes it right adjoint to $(-) \otimes A$, that is to $(-) \times A$. Hence **LinSet** is cartesian closed. $\square$

Models of intuitionistic linear logic are not generally complete and cocomplete: the logic asks for a symmetric monoidal closed category with a linear exponential modality, and the standard models — coherence spaces, phase spaces, games, and the relational construction this paper varies — supply that and no more. Here they are all present, computed on carriers, and present for one reason rather than four calculations: the forgetful functor is topological, and a

topological functor lifts every limit and colimit. So **LinSet** is at once a model of intuitionistic linear logic and a well-behaved category in the ordinary sense. Section 5 is where we would like that to matter.

5 Adding conditions to the objects

The structure of the previous sections was inherited from L and imposed nothing on the α maps: an object is a set with an arbitrary map $A \times A \rightarrow L$, and every result so far holds for all such functions. The tradition this construction comes from does impose conditions. Fourman–Scott and Higgs ask an Ω -valued set for symmetry and transitivity, drop reflexivity, and read $\alpha(a, a)$ as the extent to which a exists; their conditions are standard. Past it there is no agreed definition: sets valued in a quantale are asked for symmetry, or for an involution on the quantale, or for strength, integrality and commutativity of it, according to what a given paper needs [16]. Alvim, Mendes and Mariano [2, 3] ask four such conditions of their Q -sets. Their conditions are one point in a space the subject has not settled rather than the accepted definition, which is part of what makes it worth asking what they cost. This section asks what those conditions become over a lineale.

Two readings of α are in play and it is worth keeping them apart. Off the diagonal, $\alpha(x, y)$ is a *similarity*, the degree to which x and y are identified; on the diagonal, $E_x := \alpha(x, x)$ is an *extent*, the degree to which x exists.

Definition 5.1 (conditions on objects). A lineale-valued set (X, α) is called

- (S) *symmetric* if $\alpha(x, y) = \alpha(y, x)$;
- (T) *transitive* if $\alpha(x, y) \cdot \alpha(y, z) \leq \alpha(x, z)$;
- (B) *bounded* if $\alpha(x, y) \leq E_x$;
- (LI) *locally identical* if $E_x \cdot \alpha(x, y) = \alpha(x, y)$.

Call a condition on α *linear* if it can be written as an entailment in which each value on the left is used once, so that imposing it grants no structural rule on the values. Three of the four pass. (S) is an equality between single values, with nothing to duplicate or discard; (T) has $\alpha(x, y)$ and $\alpha(y, z)$ on the left, each once; (B) has $\alpha(x, y)$ alone. (LI) does not: an equality between a value and a product containing that value is two entailments, and each is a structural rule on the extra factor, since $\alpha(x, y) \leq E_x \cdot \alpha(x, y)$ is contraction and $E_x \cdot \alpha(x, y) \leq \alpha(x, y)$ is weakening.

Passing that test as written is not the same as being linear once instantiated, and only (S) survives the difference. (T) repeats no value at pairwise distinct arguments, but on the diagonal it does: at (x, x, y) it reads $E_x \cdot \alpha(x, y) \leq \alpha(x, y)$, the weakening half of (LI), and at (x, y, x) , using (S), it gives $\alpha(x, y) \cdot \alpha(x, y) \leq E_x$. So transitivity concedes weakening on the extent, and bounds the square of a similarity by it, without having been asked for either. (B) asks for the unsquared version of that second inequality, and is a theorem rather than an axiom twice over: when $\cdot$ is idempotent the two coincide, so (B) follows from (S) and (T), which is why the older Ω -set literature [15] asks two conditions where the lineale-valued one asks four; and when L is affine, (LI) gives $\alpha(x, y) = \alpha(x, y) \cdot E_x \leq E_x$, the derivation Alvim, Mendes and Mariano record. It is an axiom precisely when L is neither, which is our case: linear in form, cartesian in provenance.

(LI) needs no instantiation to fail. Given (T), which supplies the weakening half, it reduces to the single inequality $\alpha(x, y) \leq E_x \cdot \alpha(x, y)$ — one value on the left and two on the right, a value produced from nothing — and that inequality is the whole of what the four conditions ask beyond the linear. At $y = x$ it says $E_x \cdot E_x = E_x$. The extents must be idempotent, and idempotence is what makes an element behave as an identity: it is exactly what a lineale drops.

Proposition 5.2. *In any complete lineale L the idempotents below the unit form a locale $\text{Idm}(L)$, with $u \cdot v$ the meet and joins as in L .*

Proof. If u, v are idempotent and below 1 then $u \cdot v$ is idempotent by commutativity, and $u \cdot v \leq u \cdot 1 = u$ and likewise $\leq v$. If w is idempotent, below 1, and below both, then $w = w \cdot w \leq u \cdot v$, so $u \cdot v$ is the greatest lower bound of u and v in $\text{Idm}(L)$. For a family (u_i) , each $u_i = u_i \cdot u_i \leq (\bigvee_i u_i)^2$, so $\bigvee_i u_i \leq (\bigvee_i u_i)^2$, while distributing $\cdot$ over the joins gives $(\bigvee_i u_i)^2 = \bigvee_{i,j} u_i \cdot u_j \leq \bigvee_i u_i$; and $\bigvee_i u_i \leq 1$. Frame distributivity is the distribution of $\cdot$ over joins in L , which residuation supplies (Section 2.2). $\square$

An object satisfying (LI) therefore has all of its extents in $\text{Idm}(L)$: existence takes values in the localic part of L , and $\alpha(x, y)$ off the diagonal is untouched.

This does not make an object duplicable. Proposition 3.12 needs every value $\alpha(x, y)$ to be idempotent and below the unit; (LI) delivers idempotence on the diagonal alone, and says nothing about the bound. What it asserts is that *existence* is duplicable, which is weaker and more interesting: an object may have a classical notion of existence while its similarities between distinct points stay linear and unconstrained. A linear–non-linear adjunction [14] makes that same division between two categories; here it falls inside a single object. It is worth seeing one.

Example 5.3 (Classical existence, linear similarity). Let $L = \{\perp < a < 1\}$ with $\perp$ absorbing, 1 the unit and $a \cdot a = \perp$; the residuals are $a \multimap \perp = a$ and $a \multimap c = 1$ for $c \in \{a, 1\}$, together with $1 \multimap c = c$ and $\perp \multimap c = 1$. Define $!$ by $!1 = 1$ and $!a = !\perp = \perp$, which is a storage operator.

Take $X = \{x, y\}$ with $E_x = E_y = 1$ and $\alpha(x, y) = \alpha(y, x) = a$. All four conditions hold. The extents are classical: 1 is idempotent, lies below the unit, and is fixed by $!$. The similarity is not: $a \cdot a = \perp \neq a$, so a is not duplicable, and $!a = \perp \neq a$, so it is not modal either. Two points, then, which exist classically and are similar only linearly.

Asking $1 = \top$ of the lineale is asking that $x \leq 1$ for every x , which is weakening as a global structural rule: the logic of such an L is affine rather than linear, and affineness is precisely one of the two rules the construction exists to withhold. That is not a bookkeeping condition and we do not impose it. It is worth noting all the same that both standing examples are affine — in the Lawvere quantale the unit 0 is the top of the reversed order, which is why metric spaces are the standard affine model rather than a purely linear one, and every locale is affine as well.

What Proposition 5.2 asks is weaker, and is a condition on an object rather than on the logic: that the extents be idempotent. In the Lawvere quantale $\text{Idm}(L) = \{0, \infty\}$, so an object satisfying (LI) there has each point either wholly existent or not at all, and the device of cutting a subobject by lowering an extent degenerates to taking a subset. That is the price of (LI), and it is paid in existence alone: as Example 5.3 shows, the similarities are untouched.

6 Conclusions

The Schalk–de Paiva construction has relational morphisms, and this note runs it with functions instead. The objects, the lineale and the pointwise formulas are unchanged; what changes is

what one can say about the result. **Rel** is compact closed, so a construction with relational morphisms inherits structure from the structure of its base, and it is not clear from the relational presentation how much of what one sees is owed to **Rel** and how much to the map into L . Having run the construction with functions, we can separate the two, and what follows is the ledger.

What the base supplies. **Rel** has biproducts, so in the relational category the product and the coproduct are carried by the same object and given by the same formula; the additives are free there and say nothing about L . **Rel** is also self-dual and compact closed, which is what makes the relational negation a contravariant endofunctor with no hypothesis on the value object at all. With functions, none of this is available, and each of these has to be bought. **Sets** supplies the carriers and the maps between them, and nothing linear whatever.

What the lineale supplies. Everything in Section 3. The tensor is the multiplication of L transported pointwise; the internal hom is its residuation, indexed by $A \times A$; the negation is residuation against a chosen d , involutive exactly when d is dualizing (Proposition 3.7); the modality exists precisely when L carries a storage operator, and Girard’s isomorphisms hold exactly when the corresponding equations hold in L (Proposition 3.13). The duplicable/erasable objects are those whose values are idempotent and below the unit (Proposition 3.12). Each of these is a statement about L : the structure exists exactly when the lineale says it does. That is what the change of morphism shape was for. One entry in the column is not purely a statement about L , and it is worth saying so: the internal hom also spends a hypothesis on the shape of the carrier, since the meet over $A \times A$ that discharges the morphism condition has to exist. It is the one place where the passage to functions costs something rather than making something legible.

What the morphism shape supplies. The equalizer, which costs nothing, and the co-equalizer; and with them, once L is a lattice with top and bottom, all finite limits and colimits (Proposition 4.1); and, when L is complete, completeness and cocompleteness by way of a topological forgetful functor (Proposition 4.3). These are not available relationally, and the reason is structural rather than accidental: a self-dual category has its limits and colimits carried by the same constructions, so it cannot separate them. This is what functions buy, and it is more ordinary categorical structure than a model of intuitionistic linear logic usually has.

What we miss from the introduction Section 1 asked a third question, whether such objects support something playing the role that tripos-to-topos plays in the intuitionistic setting, and we should say why we leave it. The doctrine side is unproblematic. A tripos in the sense of [19] asks four things of an indexed poset: Heyting fibres, reindexing that preserves them, adjoints along reindexing satisfying Beck–Chevalley, and a generic predicate. Reading lineales for Heyting algebras, all four hold for $P(A) = \text{Hom}(A, L) = L^A$ with the pointwise order and reindexing by precomposition, whenever L is a quantale; and Frobenius, which is a theorem rather than an axiom in the cartesian case, survives as well, being the distribution of $\cdot$ over joins that residuation supplies. So a linear tripos in that sense is available and costs nothing, no modality being needed to obtain it. The notion is not new — it goes back to Seely [11] — though the neighbouring versions are not the one we want: BI-hyperdoctrines [4, 5] retain the cartesian structure the construction exists to do without, and the Lipschitz doctrines of [10] grade the substitution rule where we would mark it.

The difficulty is in the construction built on top. There an object is a set with a symmetric transitive predicate — the shape of Section 5 — and the identity morphism on such an object is the predicate itself, so the unit law for that composition is exactly (LI), which by Proposition 5.2 puts the extents in the localic part of L . The cartesian assumption is used at the first step, not at some later one, and that is why (LI) resists while (S), (T) and (B) do not: it is not a condition on the object in the way the others are, but the demand that a particular relation act as an identity, and an identity is idempotent by definition. For **LinSet** this is a remark rather than an obstacle, since identities here are functions and α is not asked to be one; but it closes the obvious route, and we make no claim about what a linear analogue of the construction should produce.

Whether the right response is to develop L -valued functional relations directly, at the cost of the universal properties of Section 4, which were proved with literal identities; or to look for conditions under which functions and functional relations agree; or to conclude that a linear completion should not be topos-shaped at all, we do not know. The test any answer must pass is the degenerate case: over a locale the objects are the Ω -sets and the completion had better be the sheaf topos.

A last comparison, which the ledger invites. The Dialectica construction over a lineale keeps two carriers, a set of witnesses and a set of counterexamples, with the map into L evaluating one against the other; **LinSet** is the extreme case in which the two are identified, leaving only the evaluation. That collapse is what moves the hypotheses. In Dialectica the quantifier in the morphism condition is absorbed into the counterexample component of the internal hom and never reaches the value object, and the additive conjunction takes its choice in a coproduct of index sets; here there is nowhere to put an index, so the same quantifiers have to be discharged inside L as meets and joins. A one-carrier construction therefore asks more of the value object than a two-carrier one, and the ledger above is one column of a comparison whose other column belongs to a different paper.

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